

CC1 μ 1 π Xp Selection and Extraction Procedure

MicroBooNE NuMI Run 1 FHC

Internal technical note — companion to NuMIInternalNoteCC1pi

Draft

Abstract

This note documents, end to end, the selection and cross-section extraction for the ν_μ charged-current single-charged-pion measurement on argon (CC1 μ 1 π Xp) with MicroBooNE Run 1 FHC NuMI data. It covers the signal definition, the samples, the full cut sequence and particle identification, the measured selection performance, and the extraction chain (background subtraction, response matrix, Wiener–SVD unfolding with the additional smearing matrix A_C , and the flux/target normalisation). All numbers are read from the code, configs, or the universe files; where a quantity is currently configured as a fake-data study rather than a data measurement this is stated explicitly.

1 Signal definition and phase space

An event is signal if, at truth level, the neutrino vertex is inside the fiducial volume (FV), the interaction is charged current, the neutrino is ν_μ (`|mc.nu.pdg|` = 14, i.e. $\nu_\mu + \bar{\nu}_\mu$), and the final state has exactly one muon, exactly one charged pion, no neutral pions, no kaons, and no heavier mesons; any number of protons is allowed (the “Xp”). The measured phase space is

$$p_\mu > 0.15 \text{ GeV}/c, \quad p_\pi > 0.175 \text{ GeV}/c, \quad \theta_{\mu\pi} < 2.6 \text{ rad.} \quad (1)$$

The same phase-space cuts are applied in `define_signal()` and `categorize_event()`, so signal and category assignment cannot disagree.

Table 1: Fiducial volume and target count. The dead z region is excluded.

Quantity	Value	Notes
x range	$10 < x < 246 \text{ cm}$	
y range	$-101 < y < 101 \text{ cm}$	
z range	$10 < z < 986 \text{ cm}$	dead region $675.1 < z < 775.1$ excluded
Net FV volume	$4.176 \times 10^7 \text{ cm}^2 \cdot \text{cm}$	
LAr density	1.3836 g/cm^3	
N_{Ar} (targets)	$8.710 \times 10^{29} \text{ nuclei}$	per-nucleus normalisation

2 Samples

Run 1 FHC, from `configs/file_properties_numi.txt`.

Table 2: Analysis samples. The **onBNB** slot currently holds GENIE fake data (the detVar central-value sample) for a closure study; the real beam-on sample is commented out.

Role	Sample	Purpose
onBNB	GENIE detVar-CV (fake data)	pseudo-data (closure)
extBNB	beam-off (Run 1+3)	cosmic/EXT background, 3 821 593 triggers
numuMC	NuMI overlay CV (GENIE)	response, efficiency, background model
dirtMC	dirt overlay	out-of-cryostat background
detVarCV+8	detector variations	detector systematics

3 Selection

3.1 Cut sequence

The cuts are applied in order; an event passes only if all are satisfied.

Table 3: The ten selection cuts (**CC1mu1piXp** selection class).

#	Cut	Description
1	software trigger	beam/software trigger
2	vertex in FV	reconstructed neutrino vertex inside the FV
3	topological score	Pandora topological score (neutrino-like)
4	muon track-like	muon candidate is track-like
5	pion contained	pion candidate is contained
6	muon gap	muon candidate not in a detector gap
7	pion gap	pion candidate not in a detector gap
8	shower cut	shower-multiplicity requirement
9	opening angle	$\theta_{\mu\pi} < 2.6$ rad
10	final multiplicity	final track-multiplicity requirement

3.2 Particle identification

Three boosted decision trees (BDTs) are applied via **TMVA::Reader**.

Table 4: PID BDTs. Inputs: three-plane Bragg likelihood ratios, LLR PID score, track score, track length, and SCE-corrected track end position (the pion reader omits track length).

Reader	Weights	Role
tmvaReader	dataset_MIP_BDT_no_len	MIP identification
tmvaReader_mu	dataset_muon_BDT	muon — <i>booked but not evaluated</i>
tmvaReader_pi	dataset_pion_BDT_no_len	pion

3.3 Selection performance

Measured on the **numuMC** overlay (CV-weighted) for the $\cos\theta_\mu$ binning; the integrated efficiency and purity are properties of the selection and are essentially observable-independent.

Table 5: Integrated selection performance (CV-weighted numuMC).

Quantity	Value
Total selected (reco)	$\approx 3.71 \times 10^3$ events
Selected signal	$\approx 2.21 \times 10^3$ events
All true signal (denominator)	$\approx 1.42 \times 10^4$ events
Purity	0.60
Efficiency	0.16

Figure 1 shows the selected reconstructed $\cos\theta_\mu$ spectrum split into MC signal and MC+EXT background, and the selection efficiency versus true $\cos\theta_\mu$.

4 Binning

Table 6: Differential observables and binning. Each true-bin count is one greater than the reco count; the extra true bin is the background (non-signal) bin.

Observable	Bins (true/reco)	Range
p_μ	23/22	0.15–3.00 GeV/ c
p_π	6/5	0.175–1.00 GeV/ c
$\cos\theta_\mu$	13/12	−1 to 1
$\cos\theta_\pi$	13/12	−1 to 1
$\theta_{\mu\pi}$	10/9	0–2.6 rad

5 Cross-section extraction

5.1 Formula

The flux-integrated differential cross section in true bin α is

$$\left. \frac{d\sigma}{dx} \right|_\alpha = \frac{1}{N_{\text{Ar}} \Phi \Delta x_\alpha} \sum_a U_{\alpha a} (D_a - B_a), \quad (2)$$

where D_a is the measured reco rate in bin a , B_a the estimated background (EXT + MC), $U_{\alpha a}$ the unfolding matrix (absorbing the reco→true mapping and the efficiency), N_{Ar} the FV target count, Φ the integrated flux, and Δx_α the bin width. Results are per argon nucleus, in 10^{-38} cm² differential in the observable.

5.2 Background subtraction and response

The measured reco data (D_a , the onBNB histogram) has the EXT+MC background subtracted, then is unfolded with the response built from the numuMC overlay. Figure 2 shows the background-subtracted spectrum (unfolding input) and the smearacceptance (response) matrix.

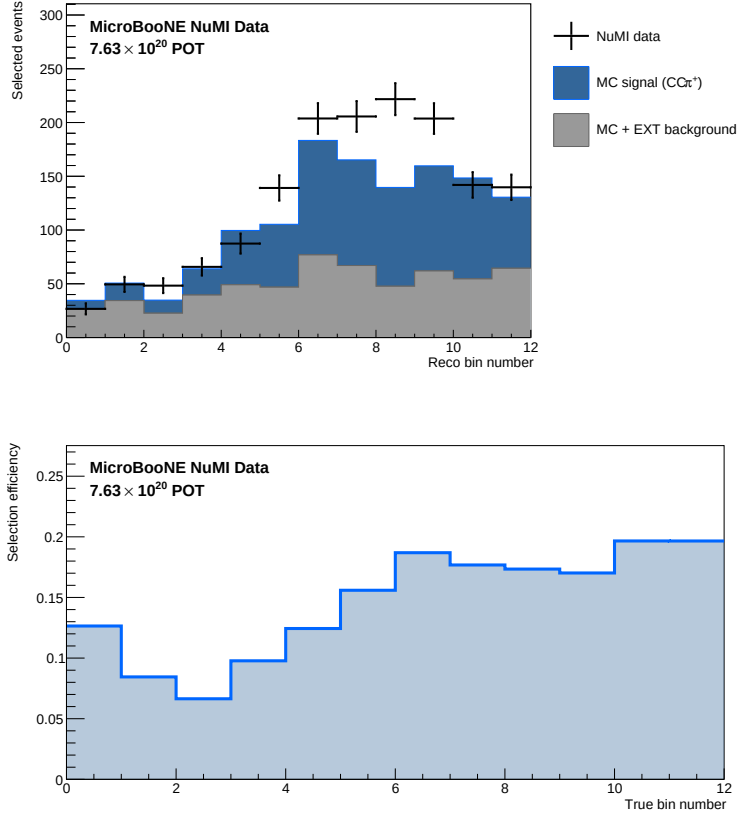


Figure 1: Top: selected reco $\cos\theta_\mu$ spectrum, data (fake data) vs MC signal + MC/EXT background. Bottom: selection efficiency vs true $\cos\theta_\mu$. The same two stages are produced for every observable.

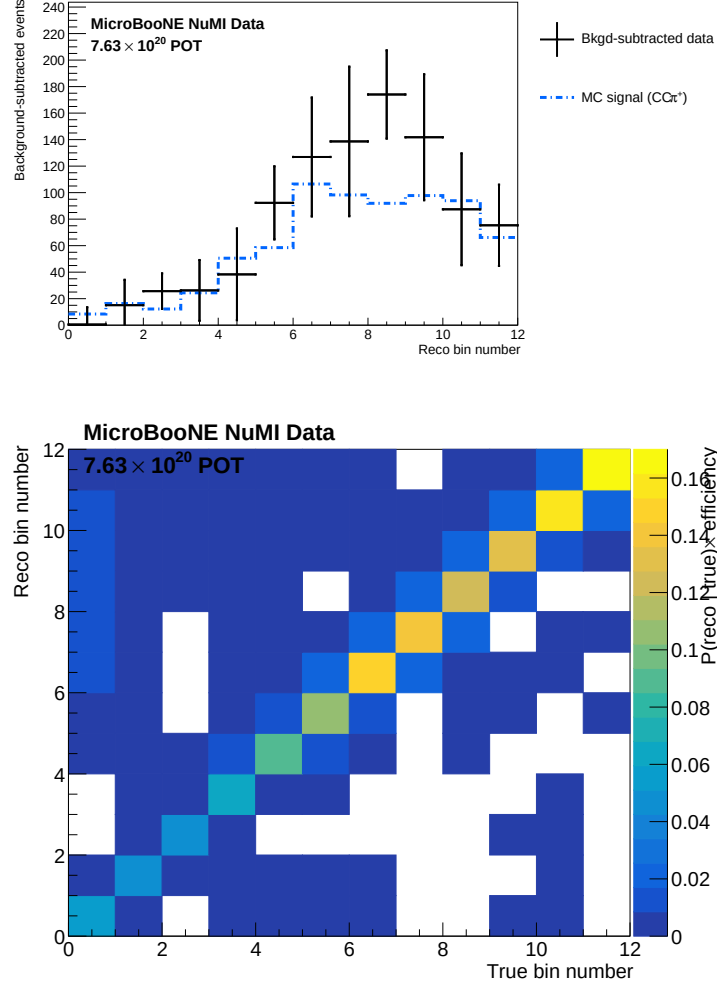


Figure 2: Top: background-subtracted reco $\cos\theta_\mu$ with the MC signal overlaid (the unfolding input). Bottom: the reco-vs-true smearacceptance (response) matrix inverted by the Wiener-SVD unfolding.

5.3 Unfolding and the additional smearing matrix

Unfolding uses Wiener–SVD with the filter enabled and second-derivative regularisation (`Unfold WienerSVD 1 second-deriv`), chosen over D’Agostini after a variant study because it suppresses the bin-to-bin oscillation from the sparse Run 1 statistics. The Wiener–SVD result carries an *additional smearing matrix* A_C : the unfolded estimator is $\hat{x} = A_C x_{\text{true}}$, so any model or truth distribution must be smeared by A_C before it is compared to the unfolded data. A_C does not in general conserve the total, so the unfolded total cross section differs from the raw truth total by the A_C normalisation; the correct closure comparison is $\hat{x}$ versus $A_C x_{\text{true}}$.

5.4 Normalisation and the conversion factor

The conversion from unfolded true signal events to cross section is $\text{conv} = N_{\text{Ar}} \Phi / 10^{-38}$, with the integrated flux $\Phi = \text{POT} \times \phi$ and the authoritative NuMI FHC flux constant $\phi = 6.81159 \times 10^{-10} \nu_\mu / \text{cm}^2 / \text{POT}$ for $E > 60 \text{ MeV}$ ($= 4.43515 \times 10^{-10} [\nu_\mu] + 2.37644 \times 10^{-10} [\bar{\nu}_\mu]$). The flux is the active-volume-averaged flux used as an approximation for the FV flux (a documented few-percent residual); no FV flux correction is applied.

Table 7: Conversion-factor inputs (fake-data POT).

Quantity	Value
N_{Ar} (FV)	8.710×10^{29}
Flux constant ϕ ($E > 60 \text{ MeV}$)	$6.81159 \times 10^{-10} \nu_\mu / \text{cm}^2 / \text{POT}$
POT	7.630913×10^{20}
Integrated flux Φ	$5.198 \times 10^{11} \nu_\mu / \text{cm}^2$
$\text{conv} = N_{\text{Ar}} \Phi / 10^{-38}$	4.527×10^3

Status note

The chain is currently configured as a fake-data study: the `onBNB` slot holds GENIE fake data and the real beam-on sample is commented out. All selection-performance and extraction numbers above are therefore validation figures, not a data measurement. See `NuMIInternalNoteCC1pi` for the fake-data closure results and the four-generator comparison.